



EE 510: Mathematical Modeling for Signal Processing

Course Code: EE 510

Course Name: Mathematical Modeling for Signal Processing

L-T-P-C: 3-0-2-4

Intended for: M.Tech EE (VLSI)

Distribution: VLSI Elective

Approval: 15th Senate (M.Tech EE-VLSI course structure, recommended by the 21st BoA); earlier listed by the 9th BoA for one-time approval as Mathematical Methods for Signal Processing (4 credits)

Course Description

In the field of signal processing we encounter problems which requires mathematical rigor – mostly in the area of linear algebra, functional analysis, optimization and estimation theory. This course develops the mathematical framework required for signal processing which consists of certain mathematical tools as well as algorithms. How these mathematical concepts are reflected in signal processing is brought out. After finishing this course the students will be in a better position to read and appreciate the current literature and problems in this area. Course will cover Linear algebra and representation in vector spaces, Linear operator theory and matrix factorization, Estimation theory, Non-linear programming/optimization.